

Layered Composition, Reciprocity, and Protocol-Relative Realization

PRINCIPIA PHYSICA Synthesis

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Abstract

This synthesis separates mathematical admissibility from physical realization. It defines a layered realized theory object, proves a reciprocity trichotomy for a linear interconnection, and proves that closed-system isomorphism need not preserve a fixed empirical protocol. It also corrects an earlier claim about limits and composition: for the oscillator family, the zero-frequency limit commutes with composition when the interaction parameter is retained. A different trajectory arises only when that interface datum is discarded. Lean and Haskell companions formalize finite logical and computational shadows of these claims; neither artifact formalizes the differential geometry in full.

1 Epistemic discipline

Assumption 1.1 (Finite-dimensional scope, S-A1). *All state spaces considered below are finite-dimensional smooth manifolds. Claims about field theory, general relativity, and quantum theory are outside the proved scope.*

Axiom 1.2 (Synthesis discipline, S-A2). *A synthesis may translate inherited claims and prove consequences from jointly satisfiable hypotheses. It may not promote an inherited epistemic label, treat agreement as evidence, or identify mathematical admissibility with empirical realization.*

Interpretation 1.3 (Founding thesis, S-I1). *The thesis that physics studies empirically realized compositional mathematical structures is treated as a methodological proposal. The mathematical results below test distinctions required by that proposal; they do not establish its exhaustiveness or empirical truth.*

2 Layered realized systems

Definition 2.1 (Realized theory object, S-D1). *A realized theory object is a tuple*

$$\mathcal{R} = (K, D, I, E, P, A),$$

where K contains states, observables, and declared geometry; D contains a generator and its flow domain; I contains ports and interaction data; E contains preparations, measurement kernels, calibration, and uncertainty; P is a protocol; and A is a predeclared adequacy rule. Mathematical admissibility concerns the appropriate mathematical components. Physical realization additionally requires empirical input in E , P , and A .

Proposition 2.2 (Vocabulary translation is not top-level injective, S-P1). *The assignment in Table 1 is a function on the listed source notions, but it is not injective into the set $\{K, D, I, E, P, A\}$. In particular, kinematical product and observable embedding both map to K , while preparation distributions and measurement kernels both map to E .*

Proof. Every row in Table 1 has one source and one target, so the displayed assignment is a function. The first two rows have the same target K , and the last two rows have the same target E . Hence the assignment is not injective into the six top-level components. The parenthetical qualifiers identify distinct substructures within those components; they do not create additional top-level tuple entries. $\square$

Source notion	Target	Internal field
Kinematical product	K	state space
Observable embedding	K	observable algebra
Dynamical factorization	D	generator and flow
Port observable	I	distinguished observable
Preparation distribution	E	preparation field
Measurement kernel	E	measurement field
Adequacy rule	A	acceptance region

Table 1: A non-injective translation into the layered carrier.

3 Reciprocity and factorization

Let $X = \mathbb{R}^{2n_1} \times \mathbb{R}^{2n_2}$ have canonical product symplectic form $\omega_\oplus$. For matrices C_{12} and C_{21} , define

$$Y = (0, -C_{12}q_2, 0, -C_{21}q_1).$$

Lemma 3.1 (Contraction calculation, S-L1). *For the stated field,*

$$\iota_Y \omega_\oplus = (C_{12}q_2) \cdot dq_1 + (C_{21}q_1) \cdot dq_2,$$

and

$$d(\iota_Y \omega_\oplus) = \sum_{a,b} ((C_{21})_{ba} - (C_{12})_{ab}) dq_{1a} \wedge dq_{2b}.$$

Proof. For a canonical pair, contraction of $(\dot{q}, \dot{p})$ with $dq \wedge dp$ is $\dot{q} dp - \dot{p} dq$. The two velocity components of Y vanish, so the dp terms drop out, and the two momentum components enter with the sign reversal carried by $-\dot{p} dq$; this gives the first display. Differentiating each coefficient and reordering $dq_{2b} \wedge dq_{1a} = -dq_{1a} \wedge dq_{2b}$ gives the second. $\square$

New result claimed by this work 3.2 (Linear composition trichotomy, S-N1). *Relative to $\omega_\oplus$, the stated family has three disjoint classes: (i) $C_{12} = C_{21} = 0$, for which the dynamics factorize; (ii) $C_{21} = C_{12}^T \neq 0$, for which the interconnection is Hamiltonian but does not factorize; and (iii) $C_{21} \neq C_{12}^T$, for which the interconnection does not preserve $\omega_\oplus$.*

Proof. By Cartan's formula and $d\omega_{\oplus} = 0$, $\mathcal{L}_Y\omega_{\oplus} = d(\iota_Y\omega_{\oplus})$. Lemma S-L1 makes this zero exactly when $C_{21} = C_{12}^T$. If both matrices vanish, $Y = 0$ and an additive factor generator remains. If the reciprocal matrices are nonzero, Y has a component depending on the other factor, so the vector field is not a product field. If reciprocity fails, the displayed Lie derivative is nonzero. These conditions are mutually exclusive and exhaustive. $\square$

Corollary 3.3 (Hamiltonian does not imply factorized, S-C1). *The decoupled class is a proper subset of the Hamiltonian class whenever a nonzero matrix C is available: take $C_{12} = C$ and $C_{21} = C^T$.*

Example 3.4 (Normal modes expose the middle class). *For one degree of freedom in each factor, take $C_{12} = C_{21} = c \neq 0$ and add identical oscillator Hamiltonians. The equations for $q_{\pm} = (q_1 \pm q_2)/\sqrt{2}$ decouple into normal modes with squared frequencies $\omega^2 \pm c/m$. Thus the total system has a Hamiltonian flow, yet neither original factor evolves without reference to the other. The example witnesses class (ii) of S-N1 without relying only on the abstract matrix criterion.*

Counterexample 3.5 (Volume preservation is not reciprocity). *The divergence of Y is zero for every pair C_{12}, C_{21} because its Jacobian has zero diagonal. Taking $C_{12} = 0$ and $C_{21} \neq 0$ therefore gives a volume-preserving interconnection in class (iii). Preservation of phase volume alone cannot replace preservation of $\omega_{\oplus}$.*

4 Closed isomorphism and empirical protocol

New result claimed by this work 4.1 (Protocol distinction survives closed isomorphism, S-N2). *Let*

$$H_m(q, p) = \frac{p^2}{2m} + \frac{m\omega_0^2 q^2}{2}.$$

For positive m, m' , the map $F(q, p) = (\alpha q, p/\alpha)$ with $\alpha = \sqrt{m/m'}$ is symplectic and satisfies $H_{m'} \circ F = H_m$. Nevertheless, a fixed protocol preparing $(q, p) = (0, 1)$ and reading the unscaled coordinate $q(t)$ distinguishes the masses whenever the two predicted readings differ.

Proof. The pullback of $dq \wedge dp$ under F is $d(\alpha q) \wedge d(p/\alpha) = dq \wedge dp$. Direct substitution gives $H_{m'}(F(q, p)) = H_m(q, p)$. The prepared solution is $q_m(t) = \sin(\omega_0 t)/(m\omega_0)$. Thus, for example, $m = 1$, $m' = 4$, $\omega_0 = 1$, and $t = 1$ give readings $\sin 1$ and $\sin 1/4$, which are unequal. The mathematical isomorphism does not transport the fixed calibration used by the protocol. $\square$

Interpretation 4.2 (Calibration transport, S-I2). *Transporting the port calibration with F removes this particular protocol difference. Keeping the calibration fixed does not. This is an interpretation of S-N2, not an empirical assertion that either oscillator is physically realized.*

Proposition 4.3 (Transported protocol restores equivalence, S-P2). *Let a protocol for H_m have preparation measure π , evolution ϕ_t , and measurement kernel K . Define the protocol for $H_{m'}$ by $\pi' = F_*\pi$, $\phi'_t = F\phi_t F^{-1}$, and $K'(x, B) = K(F^{-1}x, B)$. Then the two predicted outcome measures agree for every admitted time and event B .*

Proof. The primed prediction is

$$\int K'(\phi'_t(x), B) d\pi'(x) = \int K(F^{-1}F\phi_t(y), B) d\pi(y) = \int K(\phi_t(y), B) d\pi(y),$$

where the first equality uses the pushforward change of variables. This is the unprimed prediction. S-N2 differs because it intentionally fixes rather than transports the preparation and coordinate calibration. $\square$

Counterexample 4.4 (Partial transport is insufficient). *Transport the Hamiltonian and preparation by F but retain the original coordinate-reading kernel. The evolved mathematical states correspond, but the unscaled readout again returns $q_m(t)$ versus $q_{m'}(t)$. Dynamical intertwining alone does not establish protocol equivalence.*

5 Limits and interface retention

New result claimed by this work 5.1 (Fixed-interface commutation, S-N3). *Let $\Sigma_\omega^{(i)}$ be identical oscillators of mass m and frequency ω , and let $\otimes_\kappa$ add the interaction $\kappa q_1 q_2$. Then*

$$\lim_{\omega \rightarrow 0} (\Sigma_\omega^{(1)} \otimes_\kappa \Sigma_\omega^{(2)}) = (\lim_{\omega \rightarrow 0} \Sigma_\omega^{(1)}) \otimes_\kappa (\lim_{\omega \rightarrow 0} \Sigma_\omega^{(2)}).$$

If $\otimes_0$ is used on the right instead, its Hamiltonian differs from the fixed-interface limit by $\kappa q_1 q_2$.

Proof. Before taking the limit, the composite Hamiltonian is

$$H_{\omega, \kappa} = \frac{p_1^2 + p_2^2}{2m} + \frac{m\omega^2(q_1^2 + q_2^2)}{2} + \kappa q_1 q_2.$$

Taking $\omega \rightarrow 0$ gives $H_{0, \kappa} = (p_1^2 + p_2^2)/(2m) + \kappa q_1 q_2$. Taking the isolated limits first gives $p_i^2/(2m)$, and applying the same $\otimes_\kappa$ adds the same interaction term. Both orders therefore yield $H_{0, \kappa}$. Applying $\otimes_0$ after the factor limits instead yields $H_{0, 0}$, whose difference from $H_{0, \kappa}$ is $\kappa q_1 q_2$. $\square$

Counterexample 5.2 (Counterexample to the rejected non-commutation claim). *Retain κ in both composition operations. The two Hamiltonians are then identical. Thus a comparison using $\otimes_\kappa$ before the limit and $\otimes_0$ afterward does not establish non-commutation of a fixed operation; it establishes that changing the interface data changes the system.*

Proposition 5.3 (Vector-field form of fixed-interface commutation, S-P3). *For the Hamiltonians in S-N3, the Hamiltonian vector fields converge coefficientwise, uniformly on compact subsets of phase space, to the vector field of $H_{0, \kappa}$. Taking isolated limits and then applying the same $\otimes_\kappa$ produces that identical limiting field.*

Proof. Hamilton's equations are

$$\dot{q}_i = p_i/m, \quad \dot{p}_1 = -m\omega^2 q_1 - \kappa q_2, \quad \dot{p}_2 = -m\omega^2 q_2 - \kappa q_1.$$

On a compact set with $|q_i| \leq R$, the difference from the $\omega = 0$ field has norm at most $\sqrt{2}m\omega^2 R$. This tends uniformly to zero. Applying the retained interface after the isolated limits writes down the same last two terms, so it gives exactly the same vector field. $\square$

Interpretation 5.4 (What the estimate does not prove). *Uniform convergence of vector fields on compact phase-space sets supports finite-time trajectory comparison only together with an ODE stability argument and a common compact region containing the trajectories. It does not establish uniform agreement for all initial data or all times. The bounded-domain warning inherited from EL therefore remains relevant even though the algebraic commutation statement is exact.*

Proposition 5.5 (Finite-time trajectory estimate, S-P5). *Fix a compact convex phase-space region B and $T > 0$. Suppose the ω and zero-frequency solutions with common initial data remain in B for $0 \leq t \leq T$. Then, for a constant L bounding the Lipschitz constant of the limiting linear vector field on B ,*

$$\|z_\omega(t) - z_0(t)\| \leq \frac{\sqrt{2}m\omega^2 R}{L} (e^{Lt} - 1)$$

when $L > 0$, with the quotient replaced by t when $L = 0$.

Proof. Write the vector fields as f_ω and f_0 . The calculation in S-P3 gives $\|f_\omega(z) - f_0(z)\| \leq \varepsilon_\omega := \sqrt{2m\omega^2 R}$ on B . The integral equations imply

$$\|z_\omega(t) - z_0(t)\| \leq \int_0^t (\varepsilon_\omega + L\|z_\omega(s) - z_0(s)\|) ds.$$

Grönwall’s inequality yields the displayed estimate. The explicit compact-region hypothesis prevents the finite-time statement from being mistaken for a global uniform limit. $\square$

Counterexample 5.6 (Long times evade the estimate). *Even for a single uncoupled oscillator, $q_\omega(t) = q_0 \cos(\omega t) + (v_0/\omega) \sin(\omega t)$ approaches the free trajectory on bounded time intervals but not uniformly for all time. Taking $t = 2\pi/\omega$ with $v_0 \neq 0$ produces a discrepancy $2\pi|v_0|/\omega$. Exact commutation of the Hamiltonian formula in S-N3 therefore does not erase the domain qualifications attached to convergence of solutions.*

5.1 Protocol quotient and composition

For a fixed protocol, observational equivalence identifies realized models having identical families of predicted outcome measures. Composition can descend to this quotient only if replacing either factor by an equivalent representative leaves every composite prediction unchanged. This is a congruence requirement, stronger than equivalence of isolated predictions.

Definition 5.7 (Protocol congruence, S-D3). *An observational equivalence $\sim_P$ is a congruence for an interface-indexed operation $\otimes_\kappa$ when $M \sim_P M'$ and $N \sim_P N'$ imply $M \otimes_\kappa N \sim_{P_\kappa} M' \otimes_\kappa N'$ for the declared composite protocol P_κ .*

Counterexample 5.8 (Isolated equivalence need not be a congruence). *Let an isolated protocol read only positions at time zero. Two oscillator models with identical prepared position distributions but different momenta are equivalent under that impoverished protocol. Couple each to a second system and later read position. Hamiltonian evolution can convert the hidden momentum difference into a position difference, so the composites are distinguishable. Equivalence under the isolated protocol is therefore not automatically substitutive under richer composite interventions.*

Interpretation 5.9 (Required interface information). *The counterexample does not show that quotient composition is impossible. It shows that a protocol must expose enough port behavior, or equivalence must be defined against a sufficiently rich family of interventions, before congruence can be expected. This motivates the interface datum in S-D1 and the qualification “suitable” in S-C3.*

Empirical statement 5.10 (Computational cross-check, S-E2). *For $m = \kappa = 1$ and initial data $(0, 0, 1, 0)$, the retained-interface system has $q_1(t) = \frac{1}{2}(\cos t - \cosh t)$, while the discarded-interface system has $q_1(t) = 0$. The accompanying Haskell program evaluates both at $t = 1$ by fourth-order Runge–Kutta integration. This finite-precision calculation is a software-process record, not a proof and not an observation of a physical system.*

6 Formal scope and dependencies

Definition 6.1 (Dependency ledger, S-D2). *A dependency ledger records, for each inherited claim used by the synthesis, its source identifier, epistemic type, hypotheses, and the exact conclusion imported. Recording a dependency does not re-prove it; it makes the boundary between inherited and newly proved content inspectable.*

Proposition 6.2 (No empirical promotion by conjunction, S-P4). *Conjoining finitely many mathematical theorems and interpretations cannot by itself entail an empirical realization statement under S-A2 unless an empirical premise connecting the model to a procedure or observation is also supplied.*

Proof. S-A2 forbids identifying mathematical admissibility with realization. The conclusions of the mathematical results constrain K , D , and I , while a realization assertion additionally concerns E , P , and A by S-D1. Without a premise about those components, the required connection is absent. This is a typed consequence of the stipulated synthesis discipline, not a theorem of untyped propositional logic. $\square$

Counterexample 6.3 (Agreement of artifacts is not independent evidence). *If a Lean registry and a Haskell test are both transcribed from the same false paper premise, they may agree with one another while inheriting the same error. Their agreement checks consistency of selected encodings, not the truth of an empirical premise or the correctness of omitted differential geometry. S-E3 therefore records their limited scope explicitly.*

Empirical statement 6.4 (Tool artifact record, S-E3). *The accompanying `Proofs.lean` uses Lean core to check a finite claim registry and algebraic witnesses. The accompanying `Core.hs` provides finite numerical checks. These files do not formalize smooth manifolds, differential forms, ODE existence, or physical realization.*

Conjecture 6.5 (Interface-indexed composition, S-C3). *There may exist a category of realized theory objects in which composition is indexed by explicit interface data and is compatible with protocol quotients and suitable limits. No such category, coherence theorem, or general continuity theorem is constructed here.*

7 Bibliographic notes

The dependency identifiers CS, HR, and EL refer respectively to the supplied papers *Compositional Systems in Finite-Dimensional Classical Mechanics*, *Hamiltonian Reconstruction, Reciprocity, and Typed Physical Structure*, and *Empirical Realization, Equivalence, Limits, and Falsification*. CS-P4 is used only for the statement that isolated factors do not determine an interaction. EL-P3 and EL-P4 are cited only for their bounded-domain limit warnings. The 15 August 2026 synthesis review supplied the counterexample that forced the corrected formulation of S-N3 and identified the non-injectivity error in the former S-P1.

8 Conclusion

The synthesis establishes three scoped mathematical points. Reciprocal coupling can be Hamiltonian without being factorized. Closed Hamiltonian isomorphism does not determine equivalence under a fixed empirical protocol. For the oscillator family, a limit commutes with a fixed interaction-indexed composition, whereas discarding the interaction produces a different system. None of these mathematical results supplies evidence that a modeled structure is empirically realized.